

BAYRAMALY, Turkmenistan

WMO: 388950

Lat: 37.600N

Lon: 62.183E

Elev: 787

StdP: 14.28

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.0	22.9	6.2	7.8	22.9	12.2	10.5	31.3	16.4	57.7	14.1	53.9	4.0	80	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	105.6	71.3	103.1	70.3	100.8	69.4	74.6	99.6	72.8	98.3	71.3	96.9	5.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	96.9	92.7	63.0	88.8	89.3	61.1	82.9	87.3	38.7	99.4	37.0	98.4	35.6	97.1	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.1	11.0	9.5	13.0	109.9	7.5	2.1	7.6	111.4	3.2	112.6	-1.0	113.7	-6.4	115.2		
			11.9	77.3	7.4	2.2	6.6	79.0	2.3	80.3	-1.9	81.5	-7.3	83.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.5	40.7	43.9	54.6	65.8	77.0	85.3	88.6	85.3	74.9	62.9	50.6	42.5	(6)
(7)		DBStd	18.67	9.51	9.88	9.07	8.82	6.27	4.83	4.23	4.88	6.07	8.12	9.06	8.37	(7)
(8)		HDD50	948	308	211	57	5	0	0	0	0	0	8	99	260	(8)
(9)		HDD65	3057	752	591	337	97	4	0	0	0	5	135	438	698	(9)
(10)		CDD50	6228	21	41	200	479	836	1058	1198	1096	747	409	116	27	(10)
(11)		CDD65	2864	0	0	16	121	376	608	733	631	302	71	6	0	(11)
(12)		CDH74	38791	1	9	209	1259	4456	8760	11089	8662	3438	827	79	2	(12)
(13)		CDH80	22528	0	1	52	507	2263	5345	7105	5293	1671	271	20	0	(13)
(14)	Wind	WSAvg	4.6	5.1	5.5	5.3	4.9	4.6	4.5	4.5	4.3	4.0	4.0	4.2	4.9	(14)
(15)	Precipitation	PrecAvg	7.0	1.2	1.0	1.4	1.1	0.6	0.0	0.0	0.0	0.2	0.6	0.9	(15)	
(16)		PrecMax	10.8	4.3	4.3	3.8	3.2	2.5	0.4	0.0	0.1	0.6	1.1	3.1	2.4	(16)
(17)		PrecMin	2.8	0.0	0.1	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.8	1.0	0.8	0.7	0.9	0.6	0.1	0.0	0.1	0.3	0.6	0.7	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.9	74.6	85.9	96.2	101.4	106.6	110.1	107.6	100.7	93.2	82.8	71.1	(19)
MCWB			52.1	54.5	60.1	65.8	69.1	71.6	71.2	71.2	68.0	65.3	59.8	51.9	(20)	
2%		DB	64.3	68.5	79.6	90.5	97.5	103.5	106.5	104.2	96.7	86.5	74.9	65.4	(21)	
		MCWB	49.6	52.0	58.3	64.8	67.5	69.7	71.6	70.9	67.3	62.1	56.1	50.2	(22)	
5%		DB	59.3	63.3	74.7	86.1	94.6	101.1	103.9	101.5	93.1	82.5	69.8	60.2	(23)	
		MCWB	47.7	49.9	56.0	62.4	66.5	69.2	70.8	69.6	65.3	60.9	54.2	48.4	(24)	
10%		DB	54.0	58.7	69.7	81.5	91.2	98.8	101.2	98.6	89.4	78.5	65.4	55.5	(25)	
		MCWB	45.1	47.9	54.5	61.0	65.3	68.6	70.1	68.4	64.1	58.7	52.2	46.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	56.2	63.2	69.9	71.9	75.5	77.2	76.8	71.7	66.8	61.7	54.4	(27)
MCD B			65.7	69.6	80.7	90.0	95.5	99.5	103.0	102.3	96.1	89.1	79.6	66.7	(28)	
2%		WB	50.6	53.5	59.7	66.5	69.6	72.9	75.1	73.8	68.6	63.9	57.8	51.3	(29)	
		MCD B	61.9	65.8	75.1	85.9	93.2	98.5	99.9	99.4	92.6	83.3	70.8	62.7	(30)	
5%		WB	48.1	51.0	57.6	64.0	68.0	71.4	73.2	71.6	66.7	61.8	55.2	49.0	(31)	
		MCD B	57.4	61.3	72.0	81.4	89.9	97.2	98.9	96.9	90.2	79.8	67.6	59.1	(32)	
10%		WB	45.6	48.7	55.5	62.0	66.4	69.8	71.7	69.8	65.0	59.7	53.0	46.6	(33)	
		MCD B	53.8	57.8	68.0	78.3	87.8	95.8	97.8	95.3	87.2	76.1	64.2	54.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.8	18.5	20.2	21.9	23.8	24.3	23.9	25.5	26.4	25.2	20.7	17.0	(35)
MCD BR			25.3	26.7	28.7	29.0	28.1	27.9	27.0	28.4	30.7	30.2	27.5	25.1	(36)	
MCWBR			13.6	13.6	11.9	11.6	11.4	11.2	11.1	12.2	12.9	14.6	14.0	13.8	(37)	
5% WB		MCD BR	22.9	23.8	25.7	25.4	25.4	26.0	25.5	26.8	28.4	27.3	24.3	23.0	(38)	
		MCWBR	12.9	12.8	11.6	11.1	11.0	11.8	12.3	13.3	13.3	13.6	13.1	13.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.389	0.434	0.505	0.534	0.537	0.542	0.525	0.481	0.453	0.483	0.445	0.395	(40)	
taud		2.091	1.978	1.835	1.809	1.825	1.806	1.849	1.937	1.975	1.881	1.959	2.068	(41)		
Ebn at Noon		240	244	238	240	243	241	244	252	251	223	216	226	(42)		
Edh at Noon		38	48	61	67	67	68	64	58	52	52	43	36	(43)		
(44)	All-Sky Solar Radiation	RadAvg	711	1015	1378	1819	2256	2550	2511	2323	1918	1347	874	658	(44)	
(45)		RadStd	60	105	107	153	82	78	55	56	48	78	78	75	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-1.06	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	-1.04	-0.83	N/A	N/A	N/A	N/A

Nomenclature: See separate page